

Field Service Memo

MR

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Announcement about the tool software of log decoding for Cooling Cabinet

1. Abstract

This document announces the release of the Service tool software
"CoolingCabinetLogViewer V3.6.7.12^(*B)" of log decoding for the Cooling cabinet which is
the one of components for Elan & Orian systems, and explains how to use it.

2. Applicable system

MRT-1550

MRT-2020

Both have NO dependency to auxiliary model code (especially type of Cooling Cabinet).

3. Applicable date

The same day of this FSM issuing

4. Contents

To analyze the log of each sensor and/or position, opening the log file which is generated from Cooling Cabinet to SP, it may be required to refer the appropriate Service manual, and spend many hours trouble shooting. For the sake of reduction of analyzing load and time, we have developed the tool software “CoolingCabinetLogViewer”, which can decode the log generated from Cooling Cabinet, and displays its condition visually.

This FSM announces its release and explains how to use it below.

4.1. Purpose of this tool software

- ✓ It can decode the Cooling cabinet log, generated to SP, or downloaded to service PC from SP.
- ✓ It can assist CE's understanding of the target Cooling Cabinet's condition, by displaying the decode result visually easy-to-understand.
- ✓ This tool software is just a viewer for assistance of CE's understanding. Therefore, below functionality's are not applicable.
 - Automatic judgement about ordering applicable service parts. ^(*A)
 - Automatic/Manual downloading functionality of Cooling Cabinet log from itself or SP.
 - Other functionalities, deviating from the above purpose.

4.2. Execution environment

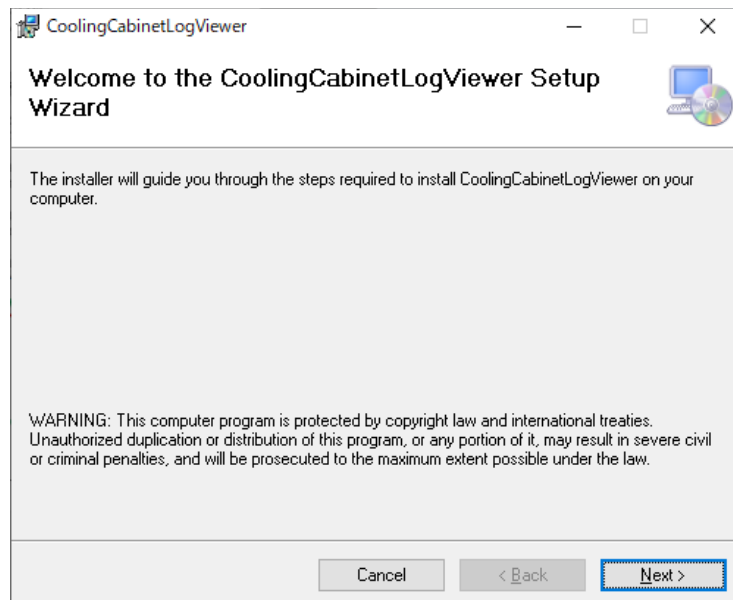
Please note to execute CoolingCabinetLogViewer under execution environment below. CoolingCabinetLogViewer has been confirmed to be able to execute and perform in service PC and SP with listed Windows and software below.

- ✓ Windows10 64bit (Japanese or English) OR Windows7 64bit / 32bit (Japanese or English)
- ✓ The latest version's LogAnalyzer, which has installed the valid license.
- ✓ An environment where all the latest Windows Update^(*A) have been applied.

4.3. Installation

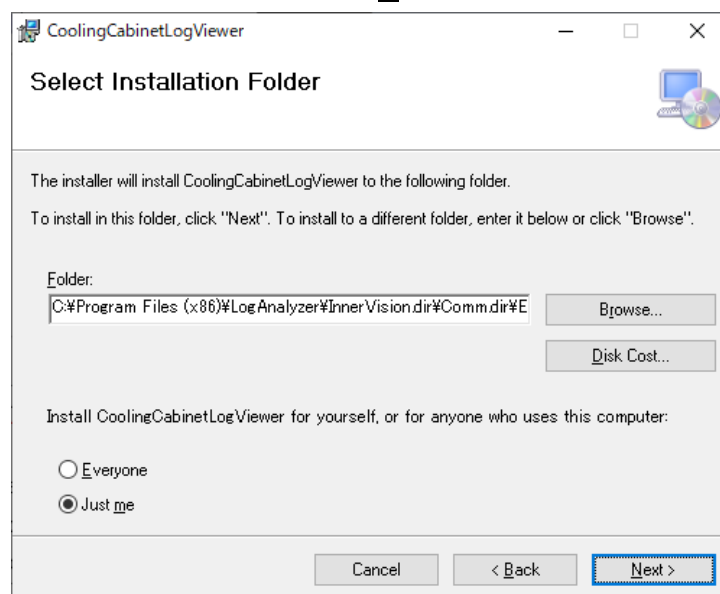
After download the Installer of CoolingCabinetLogViewer (below, CoolCabiLogViewer.msi) from FIXS knowledge, please install following below steps.

- (1) Double-click or select CoolCabiLogViewer.msi and press [Enter] key.
After showed below dialog, click [Next]^(*C) on right-below of dialog.



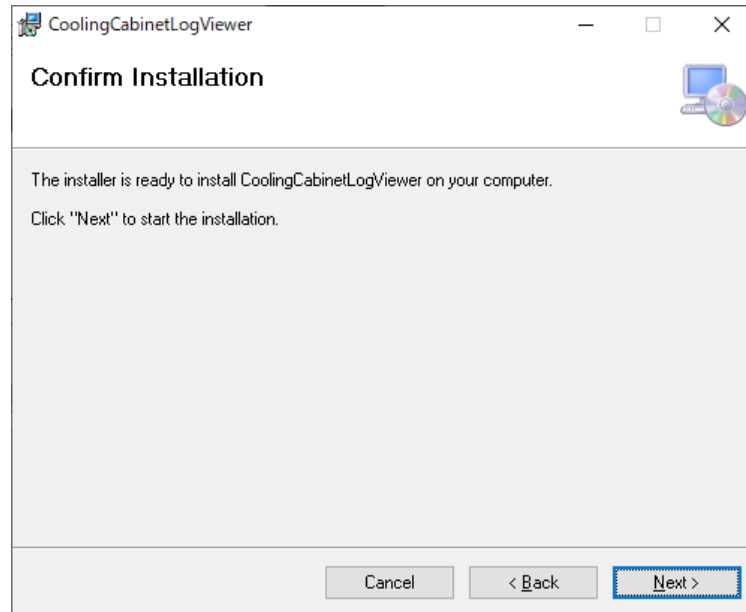
Installation entry screen of CoolingCabinetLogViewer

- (2) On "Select install folder" and selection applicable user below, **after confirmation that the same directory with installation directory of LogAnalyzer has been selected**, click [Next]^(*C) on right-below of dialog.



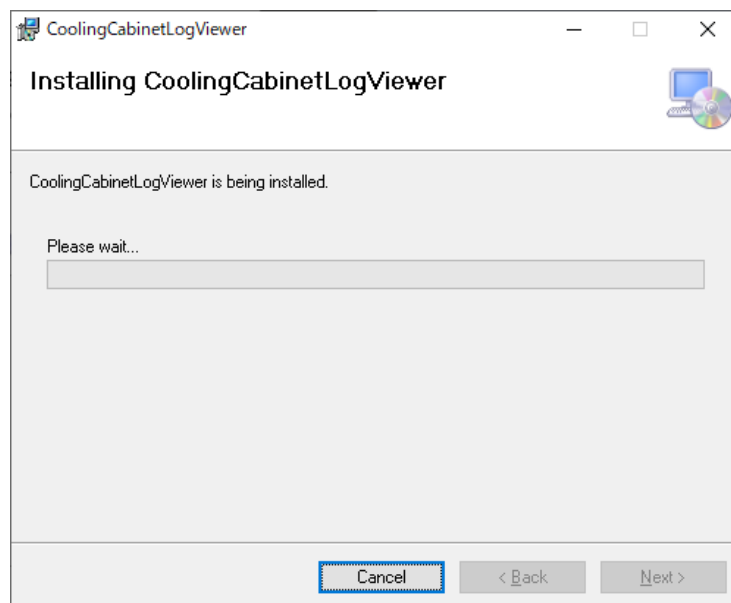
The screen of "Select install folder" and of user selection

- (3) On below screen, select [Next]^(*C) on right-below of dialog, start installation.



Confirmation screen before starting installation

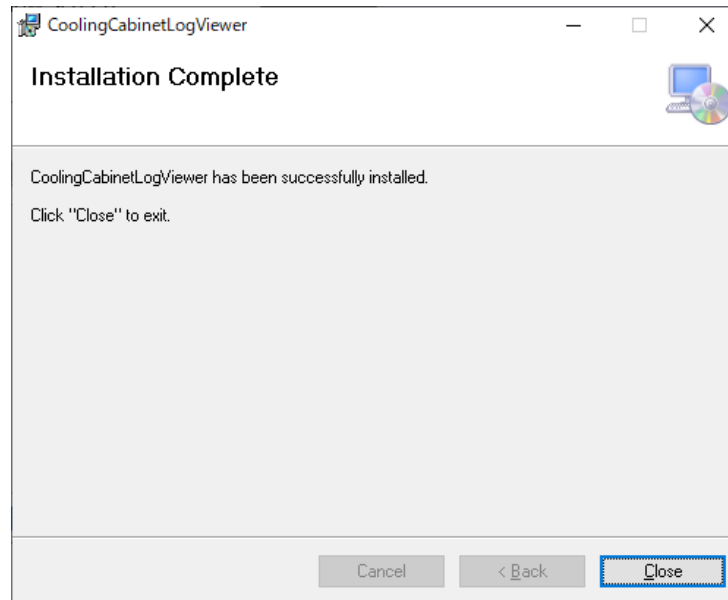
- (4) If installer starts installation, screen becomes like the example below, and progress bar under "Please wait..." message proceeds. Progress bar may be updated several times, so please wait for a while.



Installation in progress screen

- (5) If below screen is displayed, installation has been done successfully, so click [Close]^(*C) on right-below of dialog, and finish installation.

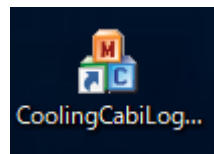
If below screen is not displayed, installation may have been failed, so restart installation steps from start (Section 4.2). If it doesn't become the screen shown below after restart installation, report phenomenon and detail steps can be found referring "5. Notes".



Installation success screen

- (6) If installation has been done successfully, short-cut icon will be generated on Desktop, invoke CoolingCabinetLogViewer from here.

Execution directly binary file pointed from short-cut is not recommended, because it may be affected on any other tool software which are placed on the same directory.



Short-cut on desktop

4.4. Invocation

CoolingCabinetLogViewer can be invoked two ways.

If the valid log file is read, discerning Elan / Orian and CCS / CCO automatically, tool software will be invoked. If tool software cannot invoke, an error description pop-up dialog may be displayed.

✓ **By selecting Cooling Cabinet log file**

If you double-click the short-cut of CoolingCabinetLogViewer, "Open file" dialog will be displayed. Select expected log file from the dialog, click [Open] on right-below of dialog.

For your information, this "Open file" dialog doesn't support Drag & Drop, although the short-cut icon of this tool software supports it.

✓ **By Drag & Drop Cooling Cabinet log file**

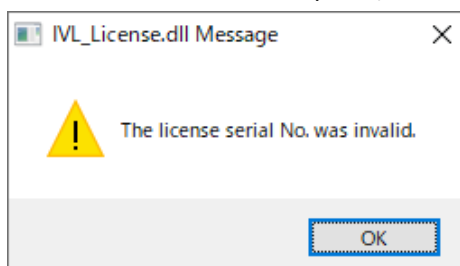
Drag & Drop the expected Cooling Cabinet log file on its short-cut.

CAUTION

- (1) Log file name opening by CoolingCabinetLogViewer should be "XXXX_CoolingCabi.txt" or "XXXX_innervisionSerial.txt"(*A). CoolingCabinetLogViewer cannot invoke(*A) without following this naming rule.
- (2) If any log format which CoolingCabinetLogViewer cannot understand is/are described in log file, the tool software cannot invoke.
- (3) If LogAnalyzer license is invalid (old, corrupted, copied, no contents, etc.…), or tool software cannot evaluate LogAnalyzer license in correct (incorrect license directory, invalid license

NOTE(*A)

In case of invalidated (ex. expired) LogAnalyzer license (below, license) existence, pop-up dialog like below will be displayed. This pop-up dialog is not shown, if invalidated licenses are deleted. And tool software can invoke, if validate license exists even only one, even if this pop-up dialog is shown.

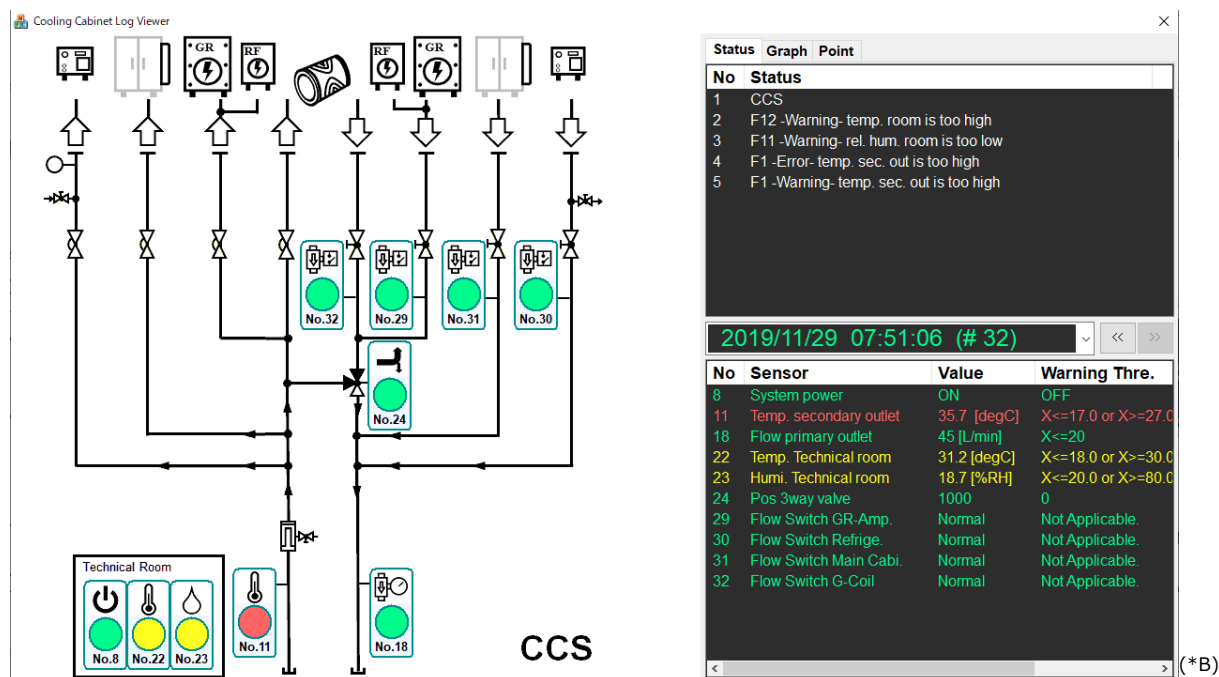


Pop-up dialog which will be displayed in case of being installed invalidated license

4.5. Running screen and operation

If CoolingCabinetLogViewer can be invoked successfully, the screen like bellows example will be displayed.

Here displays an example of reading the log file generated from CCS type Cooling Cabinet equipped in Elan. The displayed diagram and sensor contents are changed depend on MRI system type and Cooling Cabinet type.



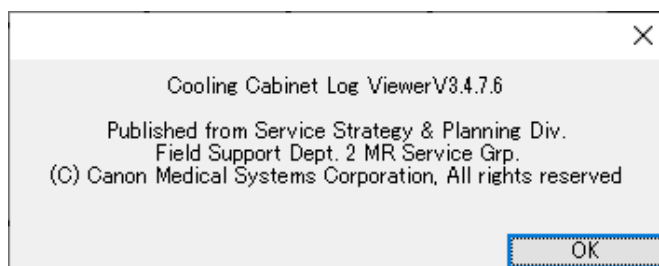
A case example of reading and invoking the log file generated from CCS type Cooling Cabinet equipped in Elan

The meaning of each part and the operation are as below.

✓ Version information^(*A)

Software version and copyright notice of invoking CoolingCabinetLogViewer is displayed in pop-up dialog.

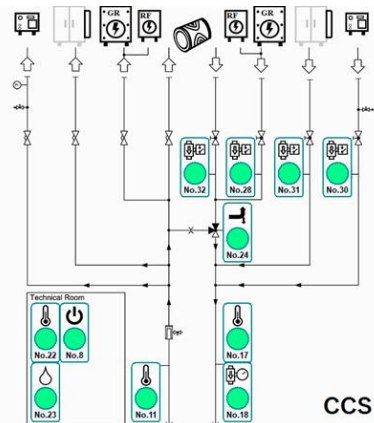
This pop-up dialog can be shown selecting "Version information Cooling Cabinet Log Viewer (&A)..." on displayed tooltip by clicking icon at left of title bar.



Version information

✓ Map (Diagram)

It displays ductwork, each part on ductwork, and flow of coolant water in Cooling Cabinet.



Map

✓ Status list

It displays Cabinet type (always) and error status (as needed). It displays only error status, if the log line which is logged only error message. (*B)

Status	Graph	Point
No	Status	
1	CCS	

Status	Graph	Point
No	Status	
1	CCS	
2	F12-Warning- temp. room is too high	
3	F11-Warning- rel. hum. room is too low	
4	F1-Error- temp. sec. out is too high	
5	F1-Warning- temp. sec. out is too high	

Status	Graph	Point
No	Status	
1	Warning - Standby mode.	

(*B)

Status list

(Left: normal, Center: an example in case of error, Right: an example in case of selecting error message line)

✓ Check point list(*A)

Check point corresponding error status displayed in Status list is displayed.

In case that, only Cabinet type is displayed (Normal), or only error status is displayed, by operating Rewind (<<) button or Forward (>>) button, nothing is displayed in Check point list.

For the sake of using as landmark showing "Confirmed", multi selection is allowed only for this list.

Status	Graph	Point
Title	Contents	

Status	Graph	Point
Title	Contents	
Error	[Dew condensation error]	
Point 1	Check humidity	
Point 2	Check Tech. room temperature	
Point 3	Check water temperature from water chiller	
Point 4	Check Air conditioner in Tech. room	
Point 5	Check temperature and humidity sensor of Tech. room	
Point 6	Replace temperature and humidity sensor	
Point 7	Replace Cooling cabinet control board	
Point 8	Replace cable in between sensor - control board	
Error	[Flow primary Out error]	

Check point list (Left: normal, Right: an example in case of error)

✓ Graph tab^(*A)

It displays graph plotting all measured value about a sensor which are selected in Sensor list. If nothing is selected in Sensor list, only vertical axis, horizontal axis, and vertical white line (below, Current line) are displayed.

It displays Measured value as dot, and grid line connected each dot. Some distance between each dot may be long, if some lines which are logged only error status exist in log file.

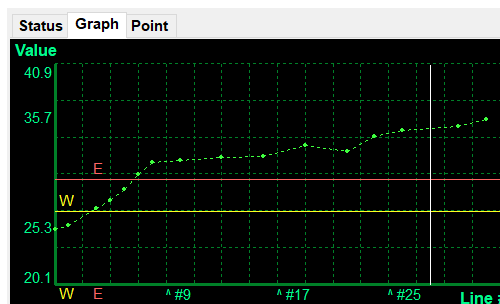
Vertical axis shows measured value, and each value means that, Maximum of Graph area, Maximum of measured value, Minimum of measured value, and Minimum of Graph area. Vertical axis is scaled automatically like Maximum and Minimum of measured value about the sensor which is selected in Sensor list are included enough in Graph area.

Horizontal axis shows line number inside of log file. 1/4, 1/2, and 3/4 of all line number are displayed as line number.

Horizontal yellow line with "W" and red line with "E" mean each threshold of the sensor which is selected in Sensor list. If threshold is/are out of graph area, only "W" and/or "E" is/are displayed at upper and/or lower end of graph area.

Current Line shows that, position inside of graph area corresponding with line number of current displayed log. It moves in operating Rewind (<<) button or Forward (>>) button.

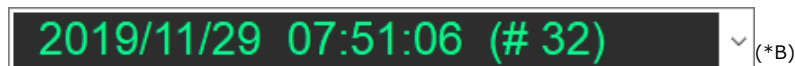
In case that Status list or Check point list is displayed and arbitrary sensor is going to be selected in Sensor list, Graph tab is shown automatically.



Graph tab (an example in case of selection arbitrary sensor in Sensor list)

✓ Date and time list

It displays current selected log date and time, and line number in log file^(*A). Clicking “v” mark on the right displays pull-down list, and enable to select any date. And mouse wheel, up key and down key, Fwd (>>) button and Rwd (<<) button, can feed date forward/backward, also. ^(*B)



Date and time list

✓ Sensor list

It displays that, each sensor name, measured value, and thresholds in string, evaluated result according to each threshold in color. If you click any column, it becomes selected condition. If you click blank area in the lowest part of list, it becomes non-selected condition.

Color means Normal as green, Warning as yellow, and Error / Alert as red.

By operating Rewind (<<) button and Forward (>>) button, if you select a log line (the day and time) logging only error status of log file, this list displays nothing.

No	Sensor	Value	Warning Thre.
8	System power	ON	OFF
11	Temp. secondary outlet	20.6 [degC]	X<=17.0 or X>=27.0
17	Temp. primary outlet	21.7 [degC]	X<=17.0 or X>=27.0
18	Flow primary outlet	69 [L/h]	X<=20
22	Temp. Technical room	24.7 [degC]	X<=18.0 or X>=30.0
23	Humi. Technical room	41.0 [%RH]	X<=20.0 or X>=80.0
24	Bypass valve position	1000	0
28	Flow Switch RF-Amp.	Normal	Not Applicable.
30	Flow Switch Refrige.	Normal	Not Applicable.
31	Flow Switch Main Cabi.	Normal	Not Applicable.
32	Flow Switch G-Coil	Normal	Not Applicable.

No	Sensor	Value	Warning Thre.
8	System power	OFF	OFF
11	Temp. secondary outlet	28.8 [degC]	X<=17.0 or X>=27.0
18	Flow primary outlet	69 [L/h]	X<=20
22	Temp. Technical room	24.9 [degC]	X<=18.0 or X>=30.0
23	Humi. Technical room	20.0 [%RH]	X<=20.0 or X>=80.0
24	Bypass valve position	500	0
29	Flow Switch GR-Amp.	Error	Not Applicable.
30	Flow Switch Refrige.	Error	Not Applicable.
31	Flow Switch Main Cabi.	Error	Not Applicable.
32	Flow Switch G-Coil	Error	Not Applicable.

Sensor list (Left: normal, Right: an example in case of error)

✓ Sensor icon

It displays position of each sensor, the evaluation result of measured value at it, sensor name, measured value, and each threshold.

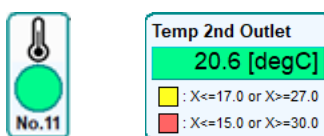
If the sensor is not selected on sensor list, corresponding sensor No., symbolic icon, and the evaluation result of measured value by color, are displayed.

If selected, each sensor's name, each threshold, measured value, and the evaluation result of measured value by color (on its edit box), are displayed.

If the log line which is logged only error message is selected, all Sensor icons disappear. (*B)

Color status indicated as:

Normal - green, Warning - yellow, and Error / Alert - red.



Sensor icon (Left: not selected, Right: selected)

✓ Rewind (<<) button

It can rewind log time of displaying. If the first row of log is selected, this button become disable.



Rewind (<<) button

✓ Forward (>>) button

It can proceed log time of displaying. If the end row of log is selected, this button become disable.



Forward (>>) button

4.6. Close

Clicking "X" button on the right-top of window can close CoolingCabinetLogViewer.

Typing "Alt + F4" is also available to close.

4.7. Uninstall

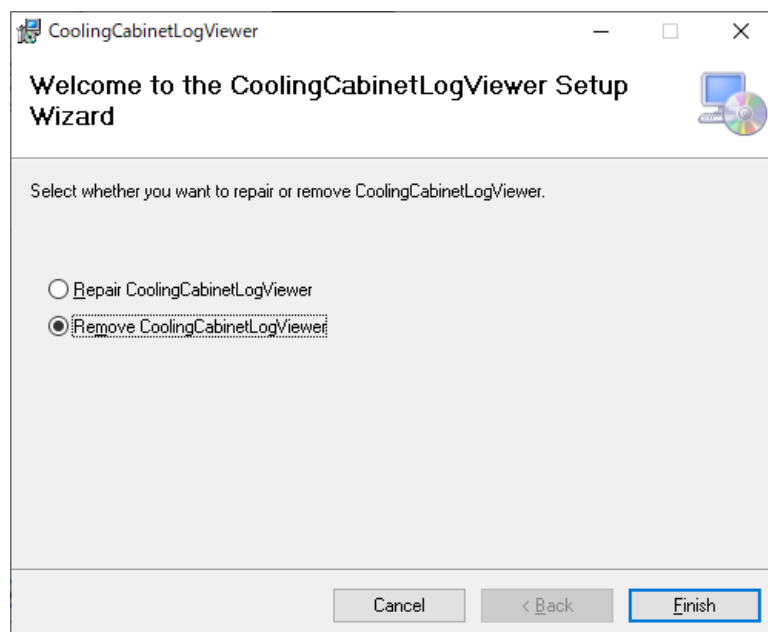
CoolingCabinetLogViewer can be uninstalled two ways.

- ✓ CoolCabiLogViewer.msi
- ✓ Windows application management

Below explain steps for each way.

✓ **Uninstall from CoolCabiLogViewer.msi**

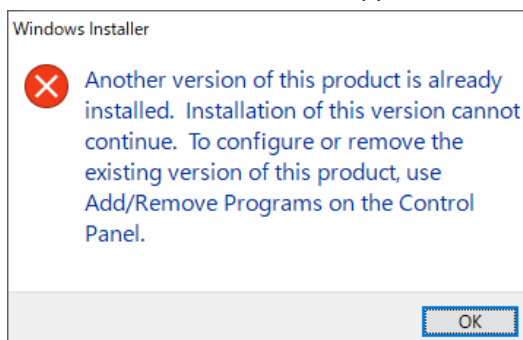
- (1) Execute CoolCabiLogViewer.msi on PC which CoolingCabinetLogViewer has already been installed, after select "Remove CoolingCabinetLogViewer" on displayed dialog, and click [Finish]^(*C) on right-bottom of dialog.



Uninstallation entry screen of CoolingCabinetLogViewer

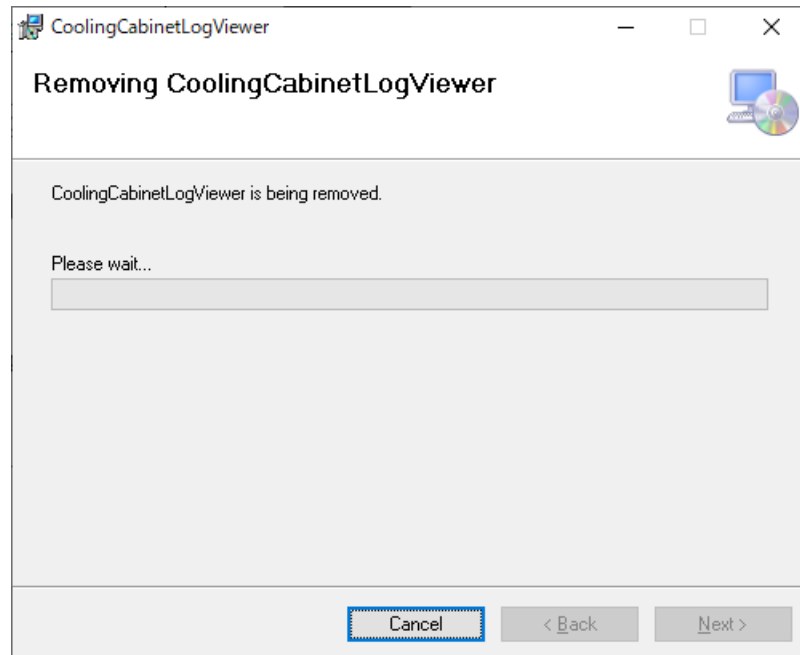
NOTE

Below dialog may be displayed, if the version between CoolCabiLogViewer.msi and installed CoolingCabinetLogViewer is different. In this case, uninstall it following steps explained "Uninstall from Windows application management".



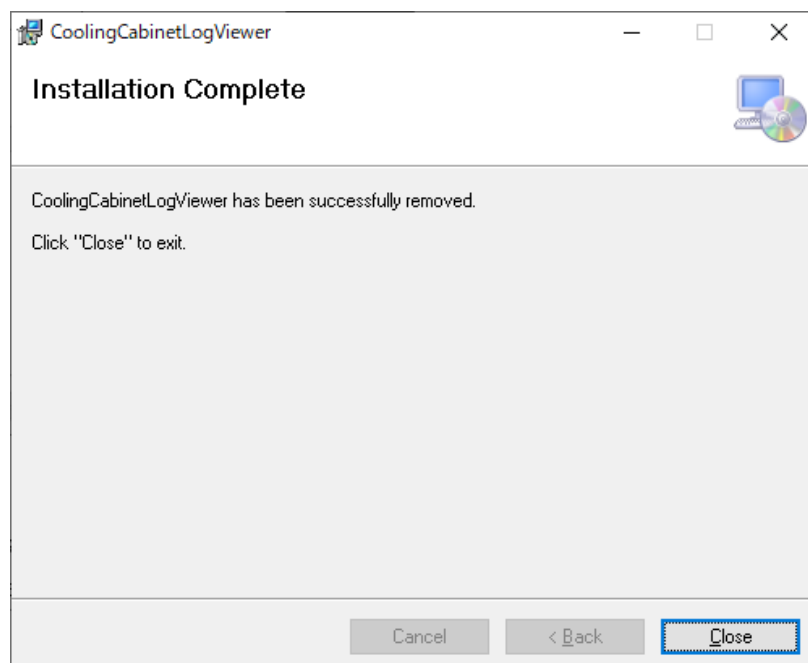
An example in case of unavailable uninstall from CoolCabiLogViewer.msi

- (2) After starting uninstallation, below screen is displayed, and progress bar under “Please wait...” message proceeds. Progress bar may be updated in several times, so please wait for a while.



Uninstallation in progress screen

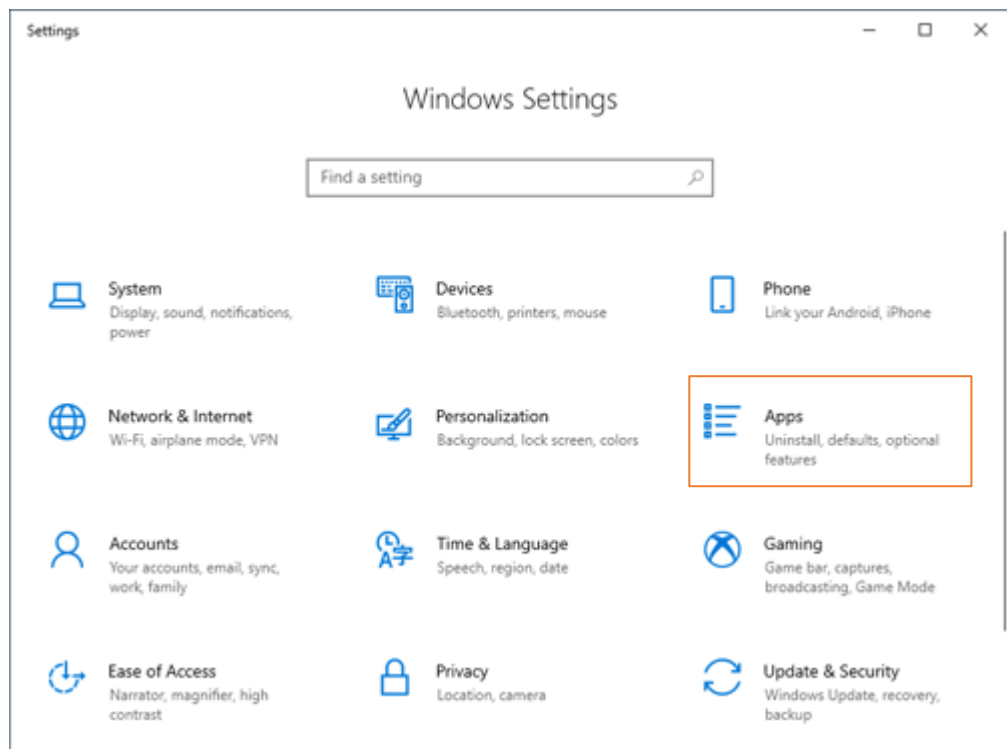
- (3) If below screen is displayed, uninstallation has been done successfully, so click [Close]^(*C) on right-below of dialog, and finish uninstallation.



Uninstallation success screen

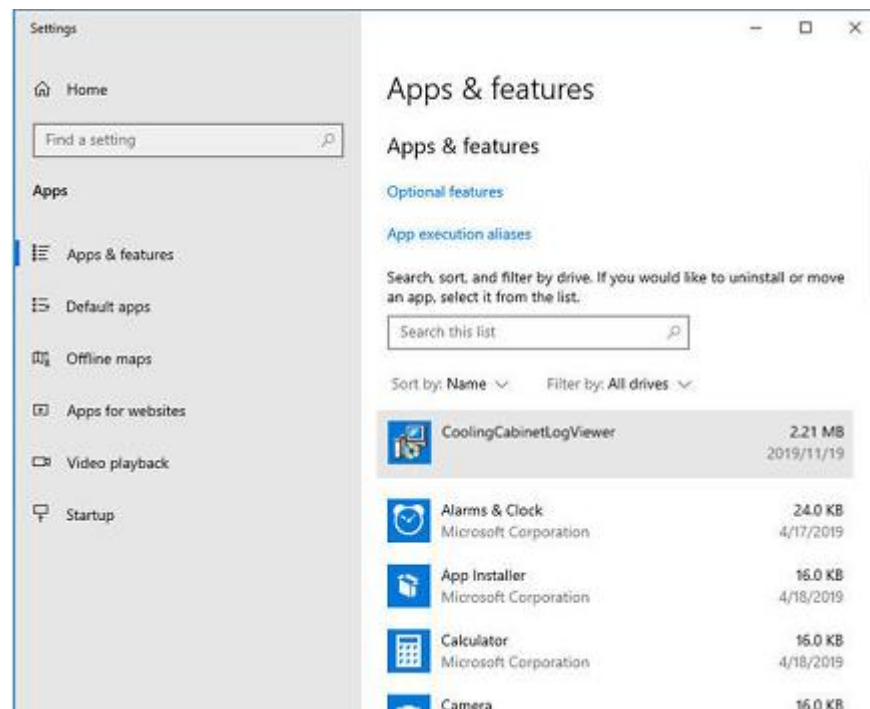
✓ **Uninstall from Windows application management**

(1) Open “Windows settings” dialog from windows start menu etc., select “Apps”.



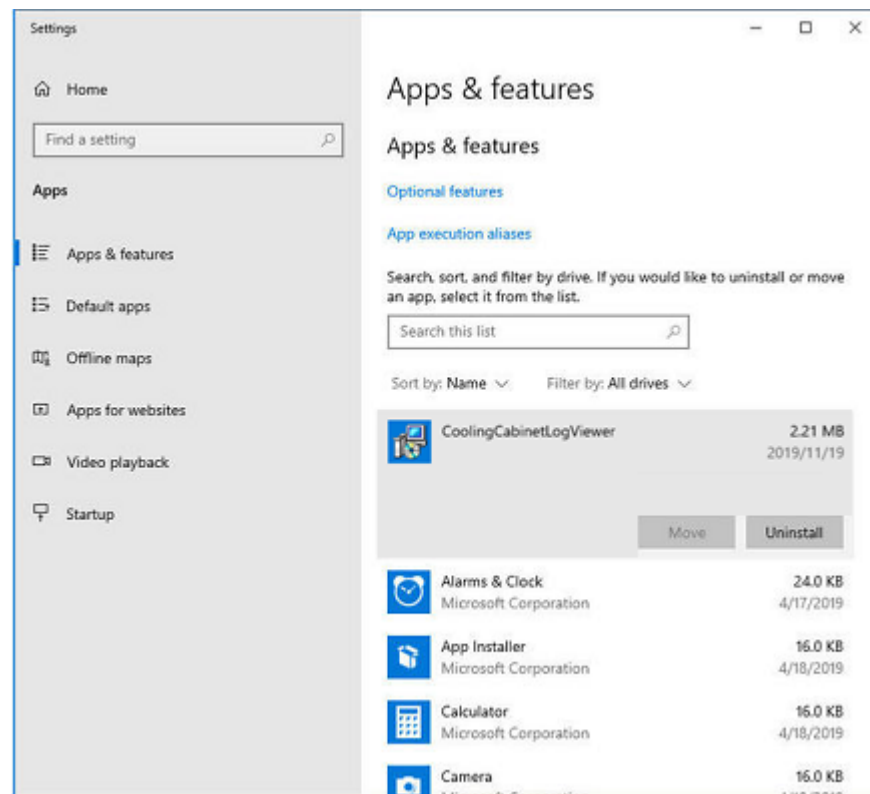
“Windows settings” screen

(2) Find and click “CoolingCabinetLogViewer” from installed application list bottom of “Apps & features” screen.



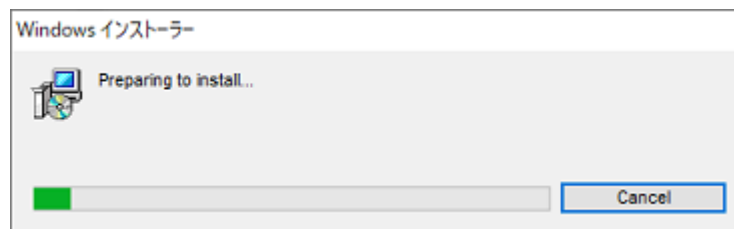
CoolingCabinetLogViewer which is displayed in installed application list

- (3) Clicking “Uninstall” inside the selected frame, a pop-up dialog is displayed, so click “Uninstall” inside the pop-up dialog.



Screen selecting “Uninstall” on CoolingCabinetLogViewer

- (4) Just after displayed pop-up dialog below, uninstall will be started, so please wait for a while. If it is completed, this pop-up dialog will disappear automatically.



Pop-up dialog displayed during execution uninstall

5. Notes

- ✓ Operation, display, and design of CoolingCabinetLogViewer may change in the future without any announcement. However, an announcement will be made in cases of major important changes.
- ✓ If In the case of a defect report or a request for improvement for is required, please contact CMSC MR Service Group.